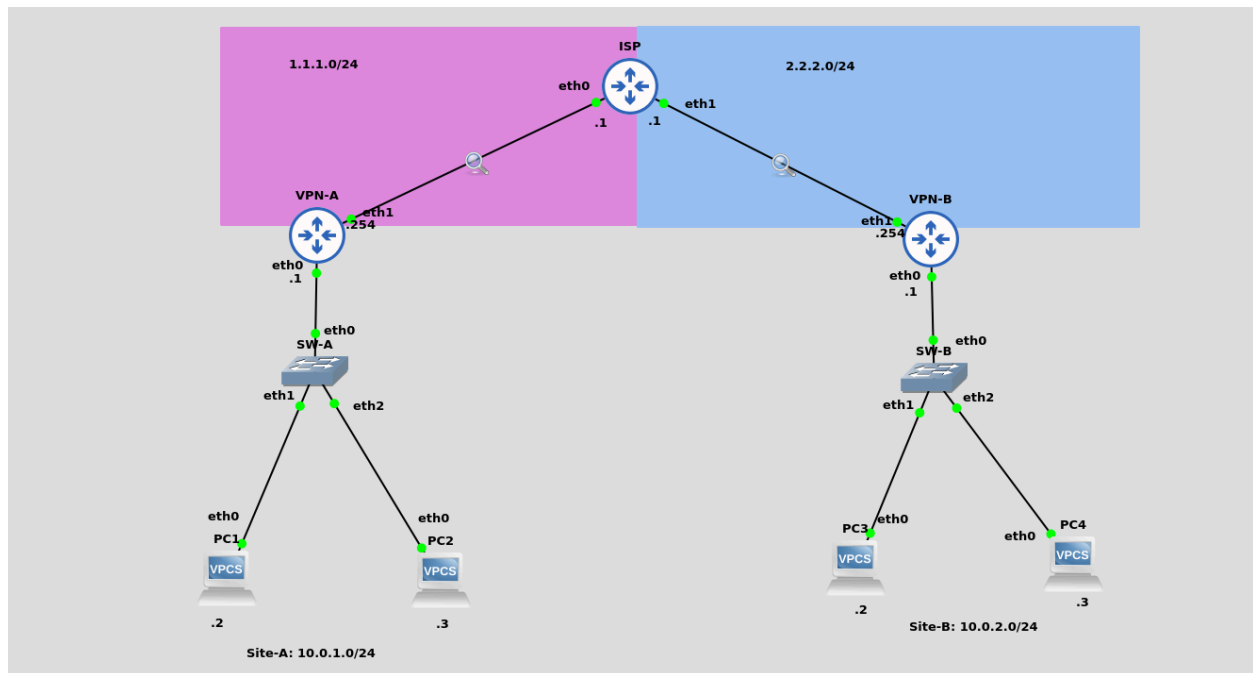


Introduction

A Virtual Private Network (VPN) extends a private network across a public network, such as the internet. It enables users to send and receive data across shared or public networks as if their devices were directly connected to the private network.

In this lab, we demonstrated this by designing two vpns (A & B) to communicate over the internet (simulated by our ISP router) using IKEv2/IPsec and WireGuard within a GNS3 (a network simulator) environment, with emphasis on packet-level verification using Wireshark (an open-source network protocol analyzer).

Topology



:
From the figure above, we have the IP addressing table of our network:

Device	Interface	IP
VPN-A (eth0)	LAN	10.0.1.1/24
VPN-A (eth1)	WAN	1.1.1.254/24
VPN-B (eth0)	LAN	10.0.2.1/24
VPN-B (eth1)	WAN	2.2.2.254/24

Why LAN/WAN separation exists?

From the table above denoting our network, we can see the LAN/WAN separation. This is due to the following reasons:

- **Trust Segmentation:**
LAN is our controlled network environment e.g our school lab in school. We have known users, servers, and controlled devices.

WAN is an untrusted environment. We can see that with the simulated ISP in our topology, outside our controlled environment

Without this boundary, internal devices would be directly reachable from outside networks, which is a major security risk.

- **Private vs Public Addressing**

LAN/WAN separation enables proper use of IP addressing:

LAN uses private IP ranges- non-routable ips. In our case, 10.0.1.0/24, 10.0.2.0/24)

WAN uses provider-assigned IPs. In our case, 2.2.2.0/24 and 1.1.1.0/24

- Routers at the LAN/WAN boundary act as decision points. In our case, VPN enforcement with IPSec/IKEv2 and WireGuard

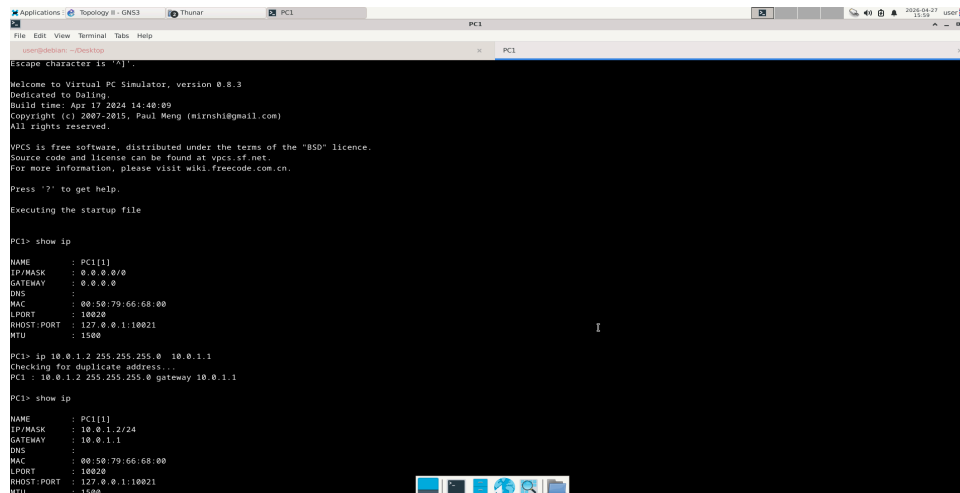
The role of our ISP router

The main reason ISP router exists in our network topology is to ensure communication between VPN A & B. The ISP router maintains a routing table and decides: “Which next hop should I use to reach this destination network?”

For IPSec/IKEv2 & WireGuard:

- The ISP router carries encrypted packets (ESP/IKE) & WireGuard
- It cannot decrypt or inspect payloads

Network Configurations



```
Applications Topology 0 - GNS3 Thunar PC1
File Edit View Terminal Tabs Help
user@debian:~$ ssh PC1
Escape character is '^['.
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Apr 17 2024 14:48:09
Copyright (c) 2007-2015, Paul Weng (minshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.
Executing the startup file

PC1> show ip
NAME      : PC1[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10020
RHOST/PORT : 127.0.0.1:10021
MTU        : 1500

PC1> ip 10.0.1.2 255.255.255.0 10.0.1.1
Checking for duplicate address...
PC1 : 10.0.1.2 255.255.255.0 gateway 10.0.1.1

PC1> show ip
NAME      : PC1[1]
IP/MASK    : 10.0.1.2/24
GATEWAY    : 10.0.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10020
RHOST/PORT : 127.0.0.1:10021
MTU        : 1500
```

This configuration above depicts the ip configuration across the VPCs in each LAN and assigning a default gateway for network traffic outside. This is to ensure each VPC can be recognised on its network

VPC example:

Ip 10.0.1.2 <host ip> 255.255.255.0 <subnet_mask> 10.0.1.1 <default gateway>

Routers' Configuration:

Earlier discussed that our routers decide where the traffic is directed to. Therefore, there is a need to configure the routers to understand their interfaces and the routing tables:

ISP Router:

Interface config:

```
ip address add address=1.1.1.1/24 interface=ether9 network=1.1.1.0
```

```
ip address add address=2.2.2.1/24 interface=ether10 network=2.2.2.0
```

Problem encountered with ISP routing table

Routing Configuration Error:

```
ip route add dst-address=10.0.1.0/24 gateway=1.1.1.1
```

```
ip route add dst-address=10.0.2.0/24 gateway=2.2.2.1
```

This erroneous configuration caused the vpn a to vpn b traffic unreachable. We are pointing the routes to the ISP router itself instead of to the next-hop routers (Site A / Site B).

Routing Configuration Fixed:

```
ip route add dst-address=10.0.1.0/24 gateway=1.1.1.254
```

```
ip route add dst-address=10.0.2.0/24 gateway=2.2.2.254
```

VPN A & B Routers:

A:

```
ip address add address=10.0.1.1/24 interface=ether9
```

```
ip address add address=1.1.1.254/24 interface=ether10
```

```
ip route add dst-address=10.0.2.0/24 gateway=1.1.1.1
```

B:

```
ip address add address=10.0.2.1/24 interface=ether9
```

```
ip address add address=2.2.2.254/24 interface=ether10
```

```
ip route add dst-address=10.0.1.0/24 gateway=2.2.2.1
```

IKEv2/IPsec

Packet Capture (Before Config)

Using Wireshark we were able to capture ICMP request and reply from VPN A to VPN B and vice versa. Before encryption, this traffic appears in plaintext on the LAN interface, showing the original source and destination IP addresses (e.g., 10.0.1.x → 10.0.2.x)

IPsec/IKEv2 Configuration

```
[admin@MikroTik] > export
# 2026-04-29 20:23:15 by RouterOS 7.14.2
# software id =
#
/interface ethernet
set [ find default-name=ether1 ] name=ether9
set [ find default-name=ether2 ] name=ether10
set [ find default-name=ether3 ] name=ether11
set [ find default-name=ether4 ] name=ether12
/ip ipsec profile
add dh-group=modp2048 enc-algorithm=aes-256 hash-algorithm=sha256 name=\
ike2-profile
/ip ipsec peer
add address=2.2.2.254/32 exchange-mode=ike2 name=vpn-b profile=ike2-profile
/ip ipsec proposal
add auth-algorithms=sha256 enc-algorithms=aes-256-cbc name=ike2-proposal \
pfs-group=modp2048
/port
set 0 name=serial0
/ip address
add address=1.1.1.254/24 interface=ether10 network=1.1.1.0
add address=10.0.1.1/24 interface=ether9 network=10.0.1.0
/ip dhcp-client
add interface=*2
/ip ipsec identity
add peer=vpn-b
/ip ipsec policy
add dst-address=10.0.2.0/24 level=unique peer=vpn-b proposal=ike2-proposal \
src-address=10.0.1.0/24 tunnel=yes
/ip route
add dst-address=2.2.2.0/24 gateway=1.1.1.1
add dst-address=10.0.2.0/24 gateway=1.1.1.1
/system note
set show-at-login=no
[admin@MikroTik] >
```

As depicted in the screenshot, this is the IKEv2/IPsec configuration in vpn A router. This same configuration exists in the vpn B router.

Basically IPsec intercepts traffic between 10.0.1.0 and 10.0.2.0 and encrypt it before sending. This is based on the following configurations on vpn A and B routers:

On VPN A/B

1) IPsec Profile (HOW the VPN behaves during negotiation): The profile defines IKE (Phase 1) behavior. It answers the question, “How do we establish and maintain the secure tunnel?”

Parameters Definition:

- name=ike2-profile
- dh-group=Diffie-Hellman group. 2048-bit key exchange
- hash-algorithm=Ensures handshake messages are not altered
- enc-algorithm=encryption used during IKE Phase 1 negotiation.

2) Create IPsec proposal (encryption settings): A proposal defines the IPsec Phase 2 (CHILD SA / ESP). This answers the question, “How should the data be encrypted and authenticated?”

- name=to reference the particular policy in case of many policies
- auth-algorithms=integrity check algorithm for ESP traffic.
- pfs-group=Diffie-Hellman group used for PFS rekeying.
- enc-algorithm=encryption algorithm used for data traffic.

3 Create IPsec Peer: An IPsec peer represents the remote VPN endpoint. In our configuration, this is the vpn-a and vpn-b for router B and A respectively.

- address=The public IP address of the remote VPN gateway
- exchange-mode=Specifies the IKE version used for key exchange
- name=Human-readable identifier for this peer
- profile=This links the peer to your IKE Phase 1 configuration

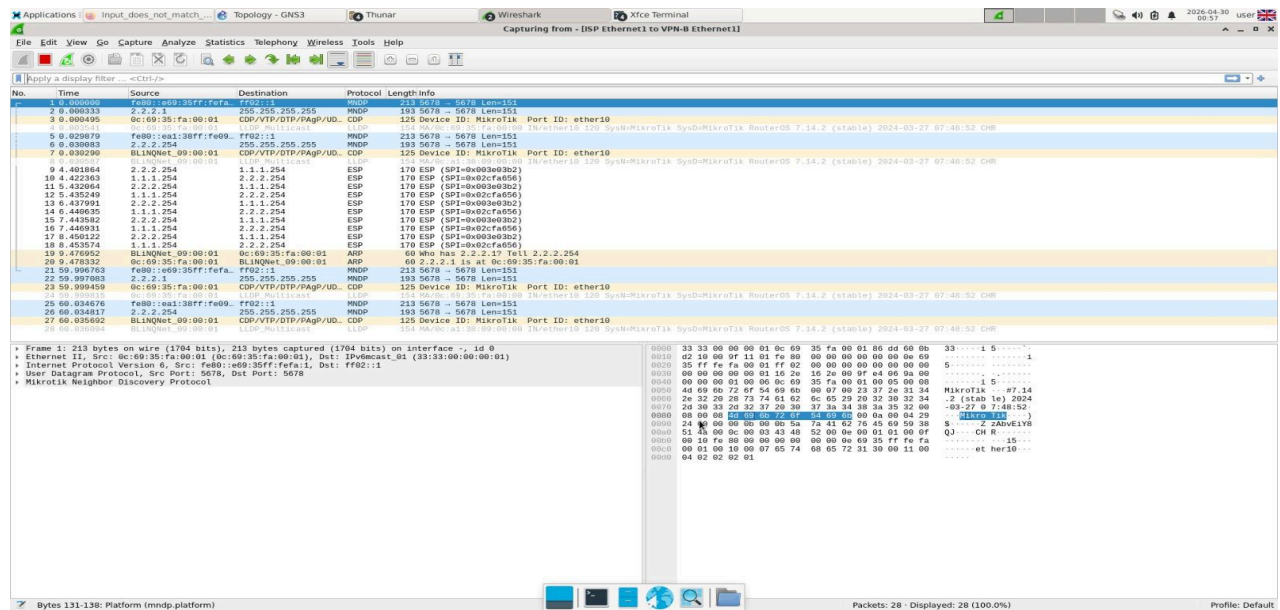
4 Create IPsec Identity: An IPsec identity is the authentication credentials binding our VPN A and B. It answers the question, “How do I prove I am allowed to build a VPN with that peer?”

- peer=We will be sharing the credential with this peer
- auth-method= In this configuration we used a pre-shared key between VPN A and VPN B
- secret= the shared secret

5 Create the actual policy: This is the part that decides what traffic goes through the VPN and how the tunnel is formed between vpn A and B.

- peer=Links this policy to the IPsec peer
- src-address=The local protected network
- dst-address=The remote protected network
- sa-src-address/ sa-dst-address= The public IP of the VPN
- tunnel=Enables site-to-site VPN mode
- level=unique. This ensures a separate IPsec SA per policy instance

Packet Capture with Wireshark after configuration: The analyzer shows the ESP (Encapsulating Security Payload)



WireGuard Configuration

```
[admin@MikroTik] > export
# 2026-04-27 19:06:47 by RouterOS 7.14.2
# software id =
#
/interface ethernet
set [ find default-name=ether1 ] name=ether9
set [ find default-name=ether2 ] name=ether10
set [ find default-name=ether3 ] name=ether11
set [ find default-name=ether4 ] name=ether12
/interface wireguard
add listen-port=13231 mtu=1420 name=wg0
/port
set 0 name=serial0
/interface wireguard peers
add allowed-address=172.16.0.1/32,10.0.1.0/24 endpoint-address=1.1.1.254 endpoint-port=13231 interface=wg0 persistent-keepalive=25s public-key="M+dKl8ONw77gkRkZzXLtQda6qquRG+cTWEasPwHi3M="
/ip address
add address=2.2.2.254/24 interface=ether10 network=2.2.2.0
add address=172.16.0.2/30 interface=wg0 network=172.16.0.0
add address=10.0.2.1/24 interface=ether9 network=10.0.2.0
/ip dhcp-client
add interface=*2
/ip route
add dst-address=10.0.1.0/24 gateway=wg0
add dst-address=1.1.1.0/24 gateway=2.2.2.1
/system note
set show-at-login=no
[admin@MikroTik] > 
```

As depicted in the screenshot, this is the WireGuard configuration in vpn B router. This same configuration exists in the vpn B router.

Create our WireGuard Tunnel Interface

```
interface wireguard add name=wg0 listen-port=13231
```

- name=Virtual tunnel interface name
- listen-port=UDP port WireGuard listens on
- By default this creates the private-key and public-key for our routers. We can share the public-key with the peer vpn but must ensure the private-key is keep private and confidential

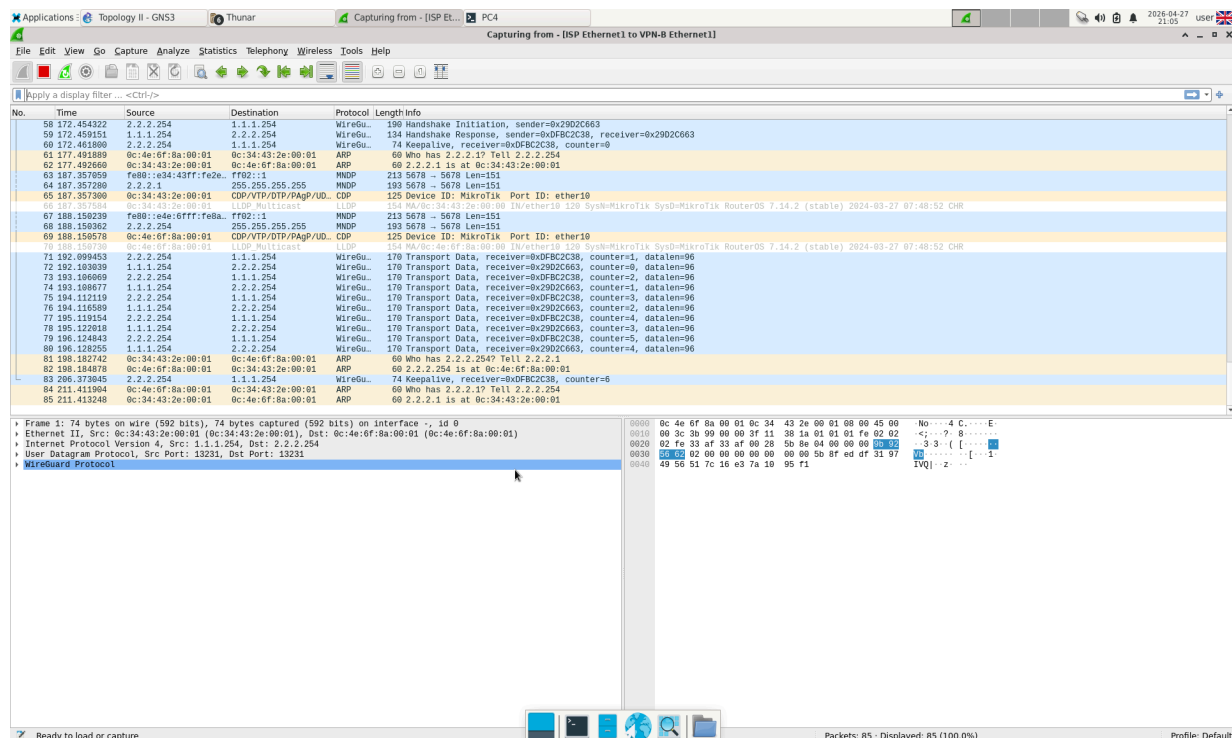
Create WireGuard Peer for each router

- Attaches this peer to WireGuard interface
- public-key=peer public key
- endpoint-address=Public IP of remote peer
- endpoint-port=UDP port of remote WireGuard service
- Allowed-address

Assigns an internal VPN IP to the interface and Route through WireGuard Interface

The lab assigned 172.16.0.0/30 for our Tunnel network:

```
ip route add dst-address= gateway=wg0
```



Lesson Learned

Challenges:

I observed the configuration challenges of the IKEv2/IPsec compared to the WireGuard configuration. This is due to different configuration steps required which is prone to human error. This was evident in my configuration with multiple attempts due to error I encountered.

Interesting Discovery

- I discovered that IKEv2/IPsec does not require tunnel interface required by our WireGuard configuration.
- As a network engineer, proper documentation cannot be overemphasized when configuring your network. It helps to avoid human error
- From the packet capture of the encrypted packet, we couldn't see the internal ip... just the isp ips

IKEv2/IPsec

Advantages

1. Strong security model. Enables different layers of security
2. More modular. I don't need to configure many tunnel interfaces

Disadvantages

1. Complex configuration
2. Heavier performance overhead. I observed different algorithms for configuration for the phases

WireGuard

Advantages

1. Very simple configuration. I only exchanged the public keys between the vpns

Disadvantages

1. Less flexible. No algorithm choice
2. Requires manual routing design

Reference:

<https://www.studocu.com/en-gb/messages/question/5956794/explain-the-difference-between-vpn-and-lan>

<https://help.mikrotik.com/docs/spaces/ROS/pages/11993097/IPsec>